

## Wiring test 1 — connection list

| from                           | to                                                     | note                              |
|--------------------------------|--------------------------------------------------------|-----------------------------------|
| Pico GP2                       | ULN2803A IN1 (pin 1)                                   | V1 fill A                         |
| Pico GP3                       | ULN2803A IN2 (pin 2)                                   | V2 vent A                         |
| Pico GP4                       | ULN2803A IN3 (pin 3)                                   | V3 fill B                         |
| Pico GP5                       | ULN2803A IN4 (pin 4)                                   | V4 vent B                         |
| ULN2803A OUT1-OUT4 (pin 18-15) | black wire V1-V4                                       | red of V1-V4 → +24 V after e-stop |
| ULN2803A COM (pin 10)          | +24 V after e-stop                                     | flyback diodes                    |
| ULN2803A GND (pin 9)           | GND (adapter minus)                                    |                                   |
| +24 V adapter                  | e-stop NC                                              | → +24 V after e-stop              |
| +24 V after e-stop             | main valve 3V210-08                                    | other side of coil → GND          |
| +24 V adapter (before e-stop)  | LM7805C IN (pin 1)                                     | 1 μF IN-GND                       |
| LM7805C OUT (pin 3)            | red wire pressure sensors A and B                      | 100 nF OUT-GND                    |
| pressure sensor A signal       | 10 kΩ → GP27, 15 kΩ GP27→GND                           | 100 nF GP27→GND                   |
| pressure sensor B signal       | 10 kΩ → GP28, 15 kΩ GP28→GND                           | 100 nF GP28→GND                   |
| potentiometer ends             | Pico 3V3 (pin 36) and GND                              |                                   |
| potentiometer wiper            | Pico GP26/ADC0                                         | 100 nF GP26→GND                   |
| all GND                        | Pico GND, ULN pin 9, 7805 pin 2, adapter minus, sensor | one common ground                 |